Tony Tran

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EXPERIENCE

Machine Learning Engineer, ISS Division June 2026 – Present
CFD Research Corporation | Huntsville, AL

- Automated detection model optimization by integrating hardware-aware Neural Architecture Search (NAS), reducing manual tuning time from 2 weeks to 1 day
- Led development of a real-time hardware-in-the-loop (HIL) simulation pipeline using C and TCP sockets, streaming FLITES-generated infrared imagery at 20 FPS and 1088x1856 resolution
- Coordinated machine learning, optics, and embedded systems teams to integrate AI models with simulated sensors and target hardware, enabling end-to-end system testing and validation

Graduate Researcher, NAIL Lab Dec 2023 – May 2026
University of Houston | Houston, TX

- Led research initiatives in efficient computer vision and edge AI, defining research objectives, designing experiments, and developing hardware-aware deep learning solutions for embedded systems
- Designed and optimized neural network search workflows reducing model search time by 59% (from 30.8 to 12.5 GPU hours)
- Accelerated ML experimentation by 3x by streamlining training and evaluation workflows on HPC clusters and Lambda Cloud, enabling faster iteration and testing

PROJECTS

ML4IRST | Real-Time Infrared Detection and Tracking

- Optimized real-time infrared object detection using TensorRT and FP16 inference on NVIDIA Jetson Orin Nano, achieving ~40 FPS for edge hardware
- Benchmarked state-of-the-art infrared detection and tracking models, identifying efficient architectures that reduced inference latency by up to 43% while maintaining competitive detection accuracy
- Improved high-resolution infrared image processing efficiency by implementing spatial-frequency decomposition in PyTorch, reducing input dimensions by 50%

ELASTIC | Hardware Efficient Object Detection on Microcontrollers

- Developed a hardware-aware Neural Architecture Search (NAS) framework using a customized Once-For-All model, achieving 72.3% mAP with 75% fewer computations than MCUNetV2
- Deployed optimized object detection models on MAX78xxx microcontrollers using hardware-aware NAS, reducing inference energy by 71.6% and latency by 2.4x compared with baseline models

OTTO | Autonomous Waste Sorting Robot

- Developed a vision-guided robotic manipulation system integrating YOLO object detection, a custom 3D-printed T-Bot gantry, and camera-to-robot coordinate mapping, achieving an 85% object pickup success rate
- Finetuned and deployed YOLOv8n on NVIDIA Jetson Nano using a custom waste detection dataset, achieving ~90% mAP and 15 FPS inference through FP16 TensorRT optimization

SKILLS

Languages: Python, C/C++

Frameworks: PyTorch, TensorFlow, TensorRT, OpenCV, Ultralytics YOLO, Detectron2, TinyEngine, NumPy, Scikit-Learn

Specializations: Computer Vision, Object Detection & Tracking, Edge AI, Model Compression and Optimization, Neural Architecture Search, Pruning & Quantization, Knowledge Distillation

Tools: Git, Linux, Docker, NVIDIA Jetson, Embedded Systems, HPC Clusters, Lambda Cloud

EDUCATION

M.S. Engineering Data Science | University of Houston 2024 – 2026

B.S. Computer Engineering Technology | University of Houston 2020 – 2024

Engineer in Training (ECE) Certified | TBPELS (#81224) 2024